SIGNED DECOMPOSITION OF THE REGRESSION STRUCTURE VIA LINEAR JEPA TRAINING: AN ALGORITHM, EMPIRICAL VALIDATION, AND A MACHINE-CHECKED PROOF

GOH CHENG ARN DAVID

Abstract. A depth-L linear Joint Embedding Predictive Architecture (JEPA) encoder, during training, assigns weight to each feature indicative of its relevance or usefulness to the prediction task. These dynamics are exactly solvable. We ask if the signed regression spectrum $\{\rho_r^*\}$ of the pencil $(\Sigma^{yx}, \Sigma^{xx})$ can be recovered from the trajectories of these weights alone, without knowing the population covariances of the features. We present an estimator whose form depends on the sign of each feature's regression coefficient ρ_r^*, which is itself read off the trajectory's asymptotic shape. On the positive branch, the plateau value recovers ρ_r^* at rate $O(\varepsilon^{1/L}|\log \varepsilon|)$, given only the eigenbasis. On the negative branch, the decay rate recovers the projected covariance at the same order, while the sample covariance recovers the quadratic form of the feature's eigenvector. The construction is packaged as the algorithm PlateauRecover. It is verified in Lean 4, with zero `sorry` and two named axioms. The ordering theorem of Goh [6] falls out as a corollary of the positive-branch trichotomy.

1. Introduction

Joint Embedding Predictive Architecture (JEPA) has been the subject of much active development. It predicts in latent/embedding space rather than reconstructing pixels, discarding irrelevant or unpredictable detail to focus on high-level structure. However, almost nothing is known rigorously about what its training dynamics do to the underlying data structure.

We find that JEPA training dynamics themselves partition every feature direction by its true predictive relationship to the target. We prove that in the linear JEPA context, we get signed, per-feature regression, decodable from the encoder's internal weight trajectory. We have an explicit signed estimator computable from the trajectory alone, and an easy to deploy algorithm we call PlateauRecover (code provided). Concretely: the sign of ρ_r^* is always recoverable from the trajectory $\sigma_r(t)$ alone, and on the positive branch so is its magnitude (given only the eigenbasis, no further population-covariance input needed) at rate $O(\varepsilon^{1/L}|\log \varepsilon|)$, a corollary of the results in §4 (Theorem 4.3).

In the linear case, everything we recover here is something OLS (or other gradient-descent methods) could already give us. We care about it because it is a critical stepping stone to the non-linear regime of representation learning on unlabeled data. There, there may be no clean y, no fixed Σ^{xx}-type inner product, and no normal-equations solver to fall back on. In particular, the LeWorldModel architecture of Maes et al. [12] is exactly such a case. There, the training trajectory is the only handle on the underlying regression structure: $\{\rho_r^*\}$ can be read off a JEPA training trajectory without ever forming Σ^{xx}, Σ^{yx}.

Notation. We fix notation for the covariances and the regression spectrum they define. For data pairs $(x, y) \in \mathbb{R}^d \times \mathbb{R}^d$, write $\Sigma^{xx} := \mathbb{E}[xx^\top] \succ 0$ for the input covariance and $\Sigma^{yx} := \mathbb{E}[yx^\top]$ for the target-input cross-covariance. The regression structure we recover throughout the paper is read off

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the generalized eigenproblem

$$\Sigma^{yx}v_r^* = \rho_r^*\Sigma^{xx}v_r^*.$$

Unlike an ordinary eigenvector, v_r^* is fixed only up to an arbitrary rescaling. The generalized eigenvalue ρ_r^* itself is untouched by this rescaling, since it is a property of the eigen-direction rather than of any particular representative v_r^*. What does move is the Σ^{xx}-quadratic form $\mu_r := v_r^* \cdot \Sigma^{xx}v_r^*$ of v_r^* (guaranteed > 0 since $\Sigma^{xx} \succ 0$), and with it the projected covariance $\lambda_r^* := \rho_r^*\mu_r$. This invariance makes ρ_r^* the signed regression coefficient the rest of the paper tracks. The collection $\{\rho_r^*\}_{r=1}^d$ (one signed scalar per feature direction) is the "regression spectrum" of the pencil $(\Sigma^{yx}, \Sigma^{xx})$ we are after. We package the triple (ρ_r^*, μ_r, v_r^*) into a single labeled object that fixes this scale-dependence once and for all:

Definition 1.1 (Signed generalised eigenbasis)**.** A *signed generalised eigenbasis* for the pencil $(\Sigma^{yx}, \Sigma^{xx})$ is a tuple $\{(\rho_r^*, \mu_r, v_r^*)\}_{r=1}^d$ with $\rho_r^* \in \mathbb{R}$, $v_r^* \in \mathbb{R}^d \setminus \{0\}$, $v_r^* \cdot \Sigma^{xx}v_s^* = 0$ for $r \neq s$, and

$$\Sigma^{yx}v_r^* = \rho_r^*\Sigma^{xx}v_r^*, \qquad \mu_r := v_r^* \cdot \Sigma^{xx}v_r^*, \qquad \lambda_r^* := \rho_r^*\mu_r. \tag{1}$$

$\mu_r > 0$ is *not* normalised to 1. It is the free Σ^{xx}-quadratic form of the (arbitrarily scaled) eigenvector v_r^*, matching `Basic.SignedGenEigenpair.hmu_def`. Rescaling v_r^* rescales μ_r and λ_r^* together while leaving $\rho_r^* = \lambda_r^*/\mu_r$ fixed, so (λ_r^*, μ_r) is only meaningful jointly, given a fixed scale for v_r^*. ρ_r^* itself is the scale-invariant quantity this paper recovers.

We now describe the model whose training we study: fix dimension d and depth $L \geq 2$ throughout (equal input/output dimension; the rectangular case is left to future work). A depth-L linear JEPA factors its encoder as $\bar{W} = W_L \cdots W_1$, each factor $W_a \in \mathbb{R}^{d\times d}$ kept separate (never collapsed into one matrix), plus a single unfactored predictor $V \in \mathbb{R}^{d\times d}$. This model trains to minimize the squared reconstruction error

$$\mathcal{L}(\bar{W}, V) \;=\; \tfrac{1}{2}\,\mathbb{E}\Big[\big\|y - V\bar{W}x\big\|_2^2\Big]\,.$$

This same eigenbasis is where the resulting training trajectory lives: under aligned balanced initialisation, the encoder's weight matrix $\bar{W}(t)$ stays diagonal in it for all $t \geq 0$. Its diagonal entry $\sigma_r(t) := v_r^{*\top}\bar{W}(t)v_r^*$ is the amplitude of feature r at time t. It obeys a Bernoulli-type ODE decoupled across features at leading order, stated precisely as Proposition 2.5 in §2.3 (equation (8)). Recovering $\{\rho_r^*\}$ means understanding what training under this loss does to $\bar{W}$ and V over time. This is where population gradient flow comes in.

Assumptions. We work throughout under the following standing assumptions.

(A1) **Exact population gradient flow.** Training proceeds under

$$\dot{\bar{W}} = -\nabla_{\bar{W}}\mathcal{L} \;\; \text{(shorthand for the preconditioned flow of Lemma 2.3)}, \qquad \dot{V} = -\nabla_V\mathcal{L}. \tag{2}$$

This is the theoretical idealisation; Algorithm 1 (§5.1) instead runs discretised SGD on finite samples.

(A2) **Aligned balanced initialisation.** We restrict throughout to initialisation at scale $\varepsilon \in (0, 1)$:

$$\bar{W}(0) = \varepsilon \cdot U_x^\top U_x, \qquad V(0) = \varepsilon \cdot U_yU_x^\top, \tag{3}$$

where U_x is the orthogonal basis of generalised eigenvectors of the pencil $(\Sigma^{yx}, \Sigma^{xx})$ and U_y its Σ^{xx}-orthonormal image under Σ^{yx} (Definition 1.1). The balance condition is preserved under (2) [1, 15]. Exact alignment cannot hold in practice — the eigenbasis is unknown at initialisation; every identifiability result here is stated under (3) exactly, with rate degradation under non-aligned initialisation not established.

(A3) **Strict spectral gap.** The signed values $\{\rho_r^*\}_{r=1}^d$ are strictly distinct and we order them as $\rho_1^* > \rho_2^* > \cdots > \rho_d^*$. Write $\delta_r := \min_{s\neq r}|\rho_r^* - \rho_s^*|$ for the gap and $\delta := \min_r \delta_r$ for the minimum gap. Needed only for the finite-sample chain (§4.3); single-feature, pure-trajectory results need no gap condition.

(A4) **Trajectory regularity.** There exist constants $K_1, c_w > 0$, independent of ε, such that along the JEPA trajectory on $[0, t^*_{\max}]$: (i) the encoder moves slowly, $\|\dot{\bar{W}}(t)\|_F \le K_1\,\varepsilon^2$; (ii) the diagonal amplitudes of Definition 1.1 stay bounded below, $\sigma_r(t) \ge c_w\,\varepsilon^{1/L}$. Both hold with equality up to constants at $t = 0$ under (3). We do not re-derive their propagation to every $t \in [0, t^*_{\max}]$ here; instead we take them as the standard trajectory-regularity condition that informal ODE learning-dynamics analyses typically leave implicit. This matches the hypothesis structure of the companion's own `bootstrap_consistency` (`hWbar_slow`, `hdiag_t` in `BootstrapLemmas.lean`).

Proof strategy. We analyze the global training dynamics by applying gradient flow at the factor level—to each matrix W_a individually and to V directly—rather than to the end-to-end product $\bar{W}$ as one matrix. This factorized parameterization induces the depth-dependent preconditioned flow of (2) (Lemma 2.3), yielding the characteristic Saxe-style depth exponents. The proof reduces this coupled matrix system to scalar trajectory recovery in four steps.

The predictor V relaxes on a timescale set by $\bar{W}(t)$'s smallest singular value, substantially faster than $\bar{W}$'s own evolution. Uniform invertibility of $\bar{W}\Sigma^{xx}\bar{W}^\top$ (Corollary 2.2) lets us substitute the quasi-static decoder $V \leftarrow V_{\text{qs}}(\bar{W})$, eliminating V analytically and reducing the joint $(\bar{W}, V)$ dynamics to an autonomous ODE in $\bar{W}$ alone (Proposition 2.4).

Projecting this ODE onto the generalised eigenbasis $\{v^*_r\}_{r=1}^d$ of $(\Sigma^{yx}, \Sigma^{xx})$, under aligned balanced initialisation $\bar{W}(t)$ remains diagonal up to a small off-diagonal leakage $c_{rs}(t)$, uniformly bounded by a fundamental-theorem-of-calculus and Cauchy–Schwarz argument (Lemma 2.1), yielding a system of decoupled scalar Bernoulli ODEs (Proposition 2.5). Diagonalising a Bernoulli-type ODE this way is classical, and we do not claim otherwise — it is the one technical step the whole paper leans on without re-deriving, the step at which simultaneous-diagonalisability of Σ^{xx} and Σ^{yx} would otherwise be needed. What is new is what the resulting closed form is used for: turning a purely qualitative dynamics fact (*which* feature is learned first) into an explicit, rate-optimal, signed estimator computable from the trajectory alone.

Each scalar ODE exhibits three qualitatively distinct regimes dictated by ρ^*_r's sign (Theorem 3.1), whose asymptotic shape uniquely identifies $\operatorname{sign}(\rho^*_r)$ (Corollary 3.3). On the positive branch, the plateau value yields ρ^*_r directly (Proposition 4.1); on the negative branch, late-time decay yields a projected-covariance estimate (Proposition 4.2). The two branches combine into the unified estimator (Theorem 4.3).

In practice only n samples are available: replacing the population covariances with sample estimates $(\hat{\Sigma^{yx}}, \hat{\Sigma^{xx}})$ introduces basis perturbation, bounded via matrix Bernstein concentration plus a Wedin-type generalized-eigenvector perturbation bound (Corollary 4.5). The chain is then packaged as an algorithm and validated empirically (§§5.1–5.2).

Outline. §§2.2–5.2 carry out the four-step argument sketched above, in the same order. §6 situates the result against prior work; §7 discusses the linear-to-non-linear transition and open questions. Appendix A catalogues the Lean signatures and the two named axioms; Appendix B gives proofs deferred from the main text, including the full finite-sample perturbation derivation.

2. Resolving the Population JEPA Flow into a System of ODEs

This section reduces the coupled $(\bar{W}, V)$ system driven by (2) to a single scalar ODE per feature, in three steps. A uniform off-diagonal bound (§2.1) guarantees invertibility along the trajectory; a quasi-static reduction eliminates V analytically (§2.2); and a projection onto the generalised eigenbasis gives the closed-form diagonal ODE (§2.3).

2.1. Off-diagonal control.

Before the quasi-static decoder can be defined, we need a uniform bound on how far $\bar{W}(t)$ drifts off-diagonal in the generalised eigenbasis $\{v^*_r\}$. This bound keeps $\bar{W}(t)\Sigma^{xx}\bar{W}(t)^\top$ invertible along the entire trajectory.

Lemma 2.1 (Off-diagonal bound). *Write $c_{rs}(t) := v_r^{*\top}\bar{W}(t)v_s^*$ for the off-diagonal amplitude between features $r \neq s$ (Definition 1.1); c_{rs} is not identically zero without the simultaneous-diagonalisability hypothesis of Littwin et al. [11]. Under Assumption A4(i)-(ii) and Assumption A2, for $L \geq 2$ there exists C', depending only on K_1, c_w, L, and the eigenbasis, such that*

$$|c_{rs}(t)| \;\leq\; C'\varepsilon^{1/L} \qquad \forall\, r \neq s,\; t \in [0, t^*_{\max}]. \tag{4}$$

Proof. Proof deferred to Appendix B (§B.1). □

Corollary 2.2 (Positive-definite lower bound). *Under Assumption A4(ii) and Lemma 2.1, provided the diagonal dominance condition $C'(d-1) < c_w$ holds, there exists $c_0 > 0$, depending only on $\lambda_{\min}(\Sigma^{xx})$, c_w, C', and d, such that*

$$\bar{W}(t)\Sigma^{xx}\bar{W}(t)^\top \;\succeq\; c_0\,\varepsilon^{2/L}\,I \qquad \forall t \in [0, t^*_{\max}],$$

so $\bar{W}(t)\Sigma^{xx}\bar{W}(t)^\top$ is invertible along the entire trajectory.

Proof. Proof deferred to Appendix B (§B.2). □

2.2. **Quasi-static decoder.** We first derive the preconditioned flow this factorization induces, before eliminating V below.

Lemma 2.3 (Coupled gradient-flow system). *The population JEPA loss $\mathcal{L}(\bar{W}, V) = \frac{1}{2}\mathbb{E}\|y - V\bar{W}x\|_2^2$ has gradients*

$$\nabla_V \mathcal{L} \;=\; \big(V\bar{W}\Sigma^{xx} - \bar{W}\Sigma^{yx}\big)\bar{W}^\top, \qquad \nabla_{\bar{W}}\mathcal{L} \;=\; V^\top\big(V\bar{W}\Sigma^{xx} - \bar{W}\Sigma^{yx}\big). \tag{5}$$

Under aligned-balanced initialisation (Assumption A2), gradient flow on the individual depth-L factors $\{W^a\}_{a=1}^L$ is equivalent to preconditioned *gradient flow on the end-to-end product $\bar{W} = W^L \cdots W^1$:*

$$\dot{\bar{W}} \;=\; -\sum_{a=1}^{L}\big[\bar{W}\bar{W}^\top\big]^{(a-1)/L}\big(-\nabla_{\bar{W}}\mathcal{L}\big)\big[\bar{W}^\top\bar{W}\big]^{(L-a)/L}, \tag{6}$$

which reduces, in the diagonal-amplitude basis, to the scalar preconditioner

$$P_{rs}(t) \;=\; \sum_{a=1}^{L}\sigma_r(t)^{2(L-a)/L}\,\sigma_s(t)^{2(a-1)/L}, \qquad P_{rr}(t) \;=\; L\,\sigma_r(t)^{2(L-1)/L}, \tag{7}$$

the latter producing the Saxe-style exponents throughout this paper.

Proof. Proof deferred to Appendix B (§B.3). □

The predictor $V(t)$ contracts to its instantaneous least-squares optimum on a fast timescale set by the smallest singular value of $\bar{W}(t)$. This is much faster than $\bar{W}(t)$ itself evolves under the balanced flow. Setting $\nabla_V\mathcal{L} = (V\bar{W}\Sigma^{xx} - \bar{W}\Sigma^{yx})\bar{W}^\top$ (Lemma 2.3) to zero and solving for V gives

$$V_{\rm qs}(\bar{W}) \;:=\; \bar{W}\Sigma^{yx}\bar{W}^\top(\bar{W}\Sigma^{xx}\bar{W}^\top)^{-1}$$

for the *quasi-static decoder*: the value of V that minimises $\mathcal{L}(\bar{W}, V)$ for fixed $\bar{W}$. This is valid whenever $\bar{W}\Sigma^{xx}\bar{W}^\top$ is invertible, which Corollary 2.2 guarantees along the JEPA trajectory — making the definition well-posed for every $t \in [0, t^*_{\max}]$, not just generically.

Proposition 2.4 (Quasi-static decoder, rigorous). *Under the JEPA flow (2) with initialisation (3), and given the invertibility of Corollary 2.2 that makes $V_{\rm qs}$ well-defined along the trajectory, there exists a constant $C_{\rm qs}$ depending only on L, $\|\Sigma^{xx}\|$, $\|(\Sigma^{xx})^{-1}\|$, and $\|\Sigma^{yx}\|$ such that*

$$\big\|V(t) - V_{\rm qs}(\bar{W}(t))\big\|_F \;\leq\; C_{\rm qs}\cdot\varepsilon^{2(L-1)/L} \qquad \forall t \in [0, t^*_{\max}],$$

where $t^*_{\max} = \mathrm{poly}(\varepsilon^{-1/L})$ is the horizon up to which Proposition 4.1 is stated.

Proof. Proof deferred to Appendix B (§B.4). □

2.3. **Generalised diagonal ODE.** Projecting the quasi-static ODE onto the eigenbasis gives one scalar Bernoulli-type equation per feature.

Proposition 2.5 (Generalised diagonal ODE). *Let $\sigma_r(t)$ be the diagonal amplitudes of Definition 1.1. Under Proposition 2.4 and Assumption A3,*

$$\dot{\sigma_r}(t) \;=\; L\,\mu_r\,\sigma_r(t)^{2-1/L}\left(\rho_r^* - \sigma_r(t)^L\right) \;+\; R_r(t,\varepsilon), \qquad |R_r(t,\varepsilon)| \le C_R\,\varepsilon^{(2L-1)/L}, \tag{8}$$

uniformly on $[0, t^*_{\max}]$, with C_R depending on the population covariances, L, and the spectral gap δ but not on ε.

The residual's actual structure is visible once the driving term is expanded, in two steps of different epistemic weight. This expansion also exposes the one place this proposition is not yet fully closed.

Proof sketch. Treating $\bar{W}(t)$ as exactly diagonal (correction controlled by Lemma 2.1) projects the preconditioned flow (6) onto $\dot{\sigma_r}(t) \approx -L\sigma_r(t)^{2(L-1)/L} \cdot v_r^{*\top}\nabla_{\bar{W}}\mathcal{L}\,v_r^*$. Writing $\Delta V := V - V_{\rm qs}(\bar{W})$, invertibility (Corollary 2.2) and the eigenpair relation collapse this exactly to $(Vv_r^*)^\top\Delta V\,\bar{W}\Sigma^{xx}v_r^*$, giving the Bernoulli drift $L\mu_r\sigma_r^{2-1/L}(\rho_r^* - \sigma_r^L)$ from ΔV's self-consistent leading-order size (a singular-perturbation reduction). The residual R_r is exactly two contributions: the gap in Proposition 2.4's bound on ΔV, and the off-diagonal correction (Lemma 2.1). Full derivation in the supplementary proofs document (§1), `https://github.com/davidcagoh/jepa-rho-recovery/blob/master/full-proofs/full-proofs.pdf`. □

Verification status and the one open gap in C_R's constant are in Appendix A.

3. Identifying the Sign of the Regression Coefficients

The ODE (8) extends naturally to signed $\rho_r^* \in \mathbb{R}$. The qualitative behaviour of σ_r splits into three regimes.

Theorem 3.1 (Sign trichotomy). *Under Proposition 2.5 with the bracket residual absorbed into R_r:*

(i) If $\rho_r^* > 0$, then $\sigma_r(t) \to (\rho_r^*)^{1/L}$ as $t \to \infty$, monotonically once $\sigma_r(t) > 0$.

(ii) If $\rho_r^* = 0$, then $|\sigma_r(t)| = O(\varepsilon^{1/L})$ for all $t \in [0, t^*_{\max}]$; the bracket is identically $-\sigma^L$, damping any positive growth at the natural ε-scale.

(iii) If $\rho_r^* < 0$, then $\sigma_r(t) \to 0$ monotonically; the bracket $-\sigma^L + \rho_r^* < 0$ for all $\sigma > 0$ forces strict decay.

The constants in the convergence rates depend on L, ρ_r^*, μ_r, and the residual constant C_R.

Proof sketch. The residual bound holds only on $[0, t^*_{\max}]$, so the asymptotic claims below concern the idealised reduction $\dot{\sigma_r} = L\mu_r\sigma_r^{2-1/L}(\rho_r^* - \sigma_r^L)$; the residual-aware, finite-time counterparts are Proposition 4.1 (case (i)) and Proposition 4.2 (case (iii)), and case (ii)'s own gap at this idealisation is Remark 3.2 below. *(i)* The bracket's sign flips at $\sigma_r = (\rho_r^*)^{1/L}$, giving monotone convergence to that fixed point from either side. *(ii)* The bracket is identically $-\sigma_r^L$, non-positive throughout — whether this confines σ_r against the residual R_r is exactly Remark 3.2's content. *(iii)* The bracket $\rho_r^* - \sigma_r^L$ is strictly negative for all $\sigma > 0$ when $\rho_r^* < 0$, and the (nonzero) drift magnitude at the initial scale shrinks more slowly in ε than the residual bound, uniformly in $L \ge 2$, so strict decay survives it; full exponent comparison in the supplementary proofs document (§5), `https://github.com/davidcagoh/jepa-rho-recovery/blob/master/full-proofs/full-proofs.pdf`. □

Verification status for each case, and why case (iii) is argued directly rather than cited, are in Appendix A.

Remark 3.2 (Zero-branch ceiling: residual bound insufficient at $L = 2$). Case (ii)'s residual bound is self-consistent with the claimed $O(\varepsilon^{1/L})$ ceiling for integer $L \ge 3$, but *not* at $L = 2$ — the depth used throughout §5.2: a sorry-free Lean proof (`ZeroBranchResidual.zero_branch_L2_ceiling_violation`, companion repository) constructs an admissible worst-case residual, consistent with Proposition 2.5's own bound, that forces σ_r above the ceiling within the trajectory horizon. This does *not* show the actual trajectory violates the ceiling — the real residual has structure a worst-case bound doesn't capture — only that the bound alone cannot prove case (ii) at $L = 2$. Full derivation in Appendix A.

3.1. **Sign identification.** The trichotomy above already pins down $\operatorname{sign}(\rho_r^*)$ from σ_r's asymptotics alone.

Corollary 3.3 (Sign identification, pure trajectory)**.** *Under the trichotomy of Theorem 3.1, the sign of ρ_r^* is determined by the asymptotic behaviour of σ_r:*

$$\operatorname{sign}(\rho_r^*) \;=\; \begin{cases} +1 & \text{if } \sigma_r(t) \text{ has a positive limit,} \\ 0 & \text{if } \sigma_r(t) \text{ is constant up to } O(\varepsilon^{1/L}), \\ -1 & \text{if } \sigma_r(t) \to 0. \end{cases}$$

Each case is detectable from σ_r alone with no parameter input.

Proof. Theorem 3.1's three cases are mutually exclusive and exhaustive in ρ_r^*'s sign. Each produces a distinct asymptotic signature of σ_r (a positive limit, boundedness at the initial ε-scale, or convergence to zero), so reading the sign off that signature is immediate case-by-case: $\rho_r^* > 0 \iff$ case (i) $\iff \sigma_r$ has a positive limit; $\rho_r^* = 0 \iff$ case (ii) $\iff \sigma_r$ stays $O(\varepsilon^{1/L})$; $\rho_r^* < 0 \iff$ case (iii) $\iff \sigma_r \to 0$. ☐

Lean targets in Appendix A.

4. Estimating the Regression Spectrum

4.1. **Positive-branch identifiability.** This is the substantive content of the paper. The trajectory σ_r on the positive branch determines ρ_r^* directly, at no cost beyond the eigenbasis itself.

The ODE (8) on the positive branch ($\rho_r^* > 0$) has a unique positive fixed point $\sigma_r^\infty = (\rho_r^*)^{1/L}$, attracting on $(0, \infty)$. The *plateau estimator*

$$\hat{\rho}_r^{\text{plateau}}(\varepsilon, T) \;:=\; \sigma_r(T)^L \tag{9}$$

reads ρ_r^* off directly once the trajectory has plateaued.

Proposition 4.1 (Plateau estimator)**.** *For $\rho_r^* > 0$, the plateau estimator $\hat{\rho}_r^{\text{plateau}}(\varepsilon, T)$ satisfies*

$$\left|\hat{\rho}_r^{\text{plateau}}(\varepsilon, T) - \rho_r^*\right| \;\le\; C_{\text{p}} \cdot \varepsilon^{1/L} \cdot |\log \varepsilon| \qquad \forall \varepsilon < \varepsilon_0,\ \forall T \ge T_*(\rho_r^*, \mu_r, \varepsilon), \tag{10}$$

where T_* is the trajectory-derived plateau-detection time (definable from σ_r alone, no parameter input) and the constants $\varepsilon_0, C_{\text{p}}, T_*$ depend on L, ρ_r^*, μ_r, and the residual constant C_R of Proposition 2.5.

Proof sketch. Write $u(t) := \sigma_r(t) - (\rho_r^*)^{1/L}$. Linearising the bracket at the fixed point turns (8) into $\dot{u} = -cu + R_r(t, \varepsilon) + O(u^2)$ with constant (in ε) contraction rate $c := L^2 \mu_r (\rho_r^*)^{(2L-2)/L} > 0$. A standard variation-of-constants bound gives $|u(t)| \le |u(0)| e^{-ct} + (C_R/c)\varepsilon^{(2L-1)/L}$; choosing the trajectory-derived time $T_* = \Theta(|\log \varepsilon|)$ at which the exponential transient has decayed to the target order (detectable from σ_r's rate of change alone, no parameter input) gives $|u(t)| = O(\varepsilon^{1/L}|\log \varepsilon|)$ for $t \ge T_*$, since $\varepsilon^{(2L-1)/L} = o(\varepsilon^{1/L}|\log \varepsilon|)$ for $L \ge 2$. The factorisation $a^L - b^L = (a - b)\sum_k a^k b^{L-1-k}$ then transfers this bound through $\sigma_r(T)^L$ to $\hat{\rho}_r^{\text{plateau}}$. Full computation in the supplementary proofs document (§2), `https://github.com/davidcagoh/jepa-rho-recovery/blob/master/full-proofs/full-proofs.pdf`. ☐

Verification status, and how the Lean proof's mechanism differs from the Grönwall argument in the full derivation, are in Appendix A.

4.2. **Signed recovery.** We combine the previous sections into the signed-decomposition headline of the abstract.

4.2.1. *Negative-branch rate analysis.* On the negative branch ($\rho_r^* < 0$), the dominant decay is set by the bare term $L\,\lambda_r^*\,\sigma^{2-1/L}$ ($\lambda_r^* < 0$ here). Asymptotically,

$$\sigma_r(t) \;\sim\; \big[(L-1)\,|\lambda_r^*|\,t\big]^{-L/(L-1)} \qquad \text{as } t \to \infty$$

in the idealised reduction (dropping R_r, on the same footing as Theorem 3.1's asymptotic claims). A monotone transform of the late-time trajectory therefore recovers $|\lambda_r^*|$.

Proposition 4.2 (Negative-branch λ_r^*-rate)**.** *For $\rho_r^* < 0$, there exists an estimator $\hat{\lambda}_r^{\rm neg}(\varepsilon, T)$, computable from σ_r on a window $[T/2, T]$ alone, achieving $\big|\hat{\lambda}_r^{\rm neg} - |\lambda_r^*|\big| \le C_\lambda^{\rm neg}\varepsilon^{1/L}$ for ε small and T in a trajectory-detectable window, with $C_\lambda^{\rm neg}$ depending on L, μ_r, and C_R.*

Proof sketch. Write $v(t) := \sigma_r(t)^{-(L-1)/L}$. By the chain rule and (8), $\dot{v}(t)$ is an *exact* identity — the leading $\sigma_r^{2-1/L}$ and $\sigma_r^{-(2L-1)/L}$ powers cancel exactly, leaving $(L-1)|\lambda_r^*|$ plus two lower-order corrections. Integrating over $[T/2, T]$ motivates the difference-quotient estimator $\hat{\lambda}_r^{\rm neg}(\varepsilon, T) := (\sigma_r(T)^{-(L-1)/L} - \sigma_r(T/2)^{-(L-1)/L})/((L-1)(T/2))$, computable from σ_r on $[T/2, T]$ alone. Bounding both correction terms using σ_r monotone decreasing on the negative branch (Theorem 3.1(iii)), evaluated once σ_r has settled to its Assumption A4(ii) floor (trajectory-detectable, no parameter input), gives exponents $\Theta(\mu_r\varepsilon)$ and $\Theta(\varepsilon^{(2L-1)(L-1)/L^2})$ — both exceeding $1/L$ for every $L \ge 2$ — so both terms are $O(\varepsilon^{1/L})$. Full computation in the supplementary proofs document (§3), `https://github.com/davidcagoh/jepa-rho-recovery/blob/master/full-proofs/full-proofs.pdf`. □

Verification status, and why this proof is argued directly rather than cited, are in Appendix A.

By contrast, μ_r enters the negative-branch ODE only through the subleading $\mu_r\sigma^{2-1/L+L}$ term (bracket $\rho_r^* - \sigma^L \approx \rho_r^*$ as $\sigma \to 0$), so recovering it from the trajectory needs controlling a quantity at order $|\rho_r^*|\,\varepsilon^{1/L}$ — dominated by the $O_p(n^{-1/2})$ sample-covariance floor for any fixed n as $\varepsilon \to 0$. We do not formalise this lower bound; the recommended estimator is therefore direct sample covariance, $\hat{\mu}_r^{\rm cov}$, combined with Proposition 4.2's $\hat{\lambda}_r^{\rm neg}$ via $\hat{\rho}_r = \lambda_r^*/\mu_r$.

4.2.2. *Signed decomposition theorem (headline).* We now combine both branches into one signed estimator, this paper's headline result.

Theorem 4.3 (Signed decomposition)**.** *Under Assumption A3, Proposition 2.5, and the conclusions of §§3–4.1 together with Corollary 3.3 and Proposition 4.2, the estimator*

$$\hat{\rho}_r(\varepsilon, T) \;:=\; \begin{cases} +\,\sigma_r(T)^L & \text{if } \sigma_r(T) > \eta(\varepsilon),\ \textit{(positive branch)} \\ 0 & \text{if } \sigma_r(T) \le \eta(\varepsilon) \text{ and } \sigma_r(T) \textit{ has plateaued},\ \textit{(zero branch)} \\ -\,\hat{\lambda}_r^{\rm neg}/\hat{\mu}_r^{\rm cov} & \text{if } \sigma_r(T) \le \eta(\varepsilon) \text{ and } \sigma_r(T) \textit{ is still decaying},\ \textit{(negative branch)} \end{cases} \tag{11}$$

where $\eta(\varepsilon) := \varepsilon^{1/L}|\log\varepsilon|^2$ is the trajectory-detectable threshold and "plateaued" vs. "still decaying" refers to the qualitative trichotomy of Theorem 3.1(ii)–(iii) — distinguished by comparing σ_r at two times (e.g. $T/2$ against T), not by the sign of $\sigma_r(T)$ at a single time, since $\sigma_r(t) \ge 0$ throughout on every branch (see the remark on (9) in §4.1 and Remark 4.4 below). Under this classification, the estimator recovers ρ_r^* at the rates

$$\big|\hat{\rho}_r - \rho_r^*\big| \;\le\; \begin{cases} C_+\varepsilon^{1/L}|\log\varepsilon| + O_p(n^{-1/2}/\delta) & \textit{positive,} \\ C_0\varepsilon^{1/L} & \textit{zero,} \\ C_-\varepsilon^{1/L} + O_p(n^{-1/2}/\delta) & \textit{negative} \end{cases}$$

uniformly in r, with constants depending on the population covariances, L, and the spectral gap δ.

Proof. By Corollary 3.3, σ_r's asymptotic signature determines $\mathrm{sign}(\rho_r^*)$ exactly, so (11)'s branch split selects the correct case for each r. On the positive branch, $\hat{\rho}_r = \sigma_r(T)^L$ and Proposition 4.1 gives the stated $C_+\varepsilon^{1/L}|\log\varepsilon|$ rate. On the zero branch, $\rho_r^* = 0$ exactly and $\hat{\rho}_r = 0$, so the error is bounded by Theorem 3.1(ii)'s $O(\varepsilon^{1/L})$ confinement translated through $\eta(\varepsilon)$, giving $C_0\varepsilon^{1/L}$. On the negative branch, $\hat{\rho}_r = -\hat{\lambda}_r^{\text{neg}}/\hat{\mu}_r^{\text{cov}}$; writing $\rho_r^* = -\lambda_r^*/\mu_r$ and applying the triangle inequality to the ratio (a first-order Delta-method expansion around (λ_r^*, μ_r), valid since $\mu_r > 0$ is bounded away from 0) combines Proposition 4.2's $C_\lambda^{\text{neg}}\varepsilon^{1/L}$ trajectory-side error with $\hat{\mu}_r^{\text{cov}}$'s standard $O_p(n^{-1/2})$ sample-covariance concentration, giving $C_-\varepsilon^{1/L} + O_p(n^{-1/2}/\delta)$. The $O_p(n^{-1/2}/\delta)$ terms shown for the positive and negative branches are the same basis-perturbation and covariance-concentration effects Corollary 4.5 isolates below; displaying them here already anticipates that combination rather than restating a purely population-level bound. Combining the three cases gives the stated uniform bound. □

Remark 4.4 (Open: finite-time noise-aware classifier). Theorem 4.3 classifies branches using the *qualitative* temporal trichotomy of Theorem 3.1, not yet a concrete, finite-time, noise-robust decision rule for telling "plateaued" apart from "still decaying" under residual and sample noise — a strictly harder question than it may first appear, since Remark 3.2 shows even the qualitative ceiling this classifier would rely on is unresolved at $L = 2$. Left to future work; full scope in Appendix A.

Verification status and scope are in Appendix A. The zero- and negative-branch classification rule of (11) is stated qualitatively per Remark 4.4 and is not itself part of the Lean-verified chain.

4.3. **Finite-sample rate.** The estimators above use the sample eigenbasis $\{\hat{v}_r\}$ of $(\hat{\Sigma^{yx}}, \hat{\Sigma^{xx}})$ in place of the population eigenbasis $\{v_r^*\}$. This introduces basis-perturbation error alongside Theorem 4.3's ε-dependent dynamics error.

Corollary 4.5 (End-to-end finite-sample rate)**.** *On an event of probability at least* $1 - \nu$ *under which* $\|\hat{\Sigma^{xx}} - \Sigma^{xx}\|_{\text{op}}, \|\hat{\Sigma^{yx}} - \Sigma^{yx}\|_{\text{op}} \le C''\sqrt{\log(d/\nu)/n}$ *(Appendix B, eq. (12)), the signed-decomposition estimator* $\hat{\rho}_r$ *of* (11) *computed in the sample eigenbasis* $\{\hat{v}_r\}$ *satisfies, uniformly in* r,

$$\left|\hat{\rho}_r - \rho_r^*\right| \;\le\; C \cdot \varepsilon^{1/L} |\log \varepsilon| \;+\; \frac{C'}{\delta}\sqrt{\frac{\log(d/\nu)}{n}},$$

with C, C' *depending on* L*, the population covariances, and* σ_x, σ_y*.*

Proof sketch. Writing $\hat{\rho}_r(v_r)$ for the estimator (11) read off in coordinate direction v_r, the triangle inequality splits $|\hat{\rho}_r(\hat{v}_r) - \rho_r^*|$ into a basis-perturbation term $|\hat{\rho}_r(\hat{v}_r) - \hat{\rho}_r(v_r^*)|$ and a dynamics-error term $|\hat{\rho}_r(v_r^*) - \rho_r^*|$. The second is exactly Theorem 4.3's $O(\varepsilon^{1/L}|\log \varepsilon|)$ bound. The first uses that $v_r \mapsto \hat{\rho}_r(v_r)$ is Lipschitz near v_r^* (constant bounded by $\|\bar{W}(t)\|$ along the trajectory), so $|\hat{\rho}_r(\hat{v}_r) - \hat{\rho}_r(v_r^*)| \le L_\rho \sin\angle(\hat{v}_r, v_r^*)$; matrix Bernstein concentration [18, Thm 1.6.2] plus a Wedin-type bound for the generalised eigenproblem (Appendix B) bounds this angle by $(C_W/\delta)\sqrt{\log(d/\nu)/n}$. Combining the two terms gives the stated bound with $C' := L_\rho C_W$. Full computation in the supplementary proofs document (§4), `https://github.com/davidcagoh/jepa-rho-recovery/blob/master/full-proofs/full-proofs.pdf`. □

Verification status is in Appendix A.

5. Algorithm and Empirical Validation

5.1. **Algorithm PlateauRecover.** Theorem 4.3 is constructive. Each piece of the estimator $\hat{\rho}_r$ in (11) is computable from the JEPA training trajectory and (on the negative branch) the sample covariances. We package the construction as Algorithm 1.

5.2. **Empirical validation.** We report two synthetic experiments, both restricted to the *positive branch* ($\rho_r^* > 0$). Experiment 1 sanity-checks the diagonal-amplitude ODE (8) of §2.3 against direct numerical integration of the JEPA gradient flow. Experiment 2 isolates the $\varepsilon^{1/L}|\log \varepsilon|$ dynamics-error rate of Proposition 4.1 from the $O_p(n^{-1/2})$ sample-noise floor of Corollary 4.5.

Algorithm 1 : PlateauRecover.

Input: samples $X, Y \in \mathbb{R}^{n\times d}$; depth $L \geq 2$; init scale $\varepsilon > 0$; plateau tolerance $\tau_p > 0$; horizon $T_{\max} \in \mathbb{N}$; rate $\eta > 0$.

Output: $\{\hat{\rho}_r\}_{r=1}^{d} \in \mathbb{R}^d$.

Step 1. Sample covariances $\hat{\Sigma^{xx}} = X^\top X/n$, $\hat{\Sigma^{yx}} = Y^\top X/n$.

Step 2. Generalised eigendecomposition of $(\hat{\Sigma^{yx}}, \hat{\Sigma^{xx}})$: sample eigenvectors $\{\hat{v}_r\}$, orthonormalisation $\{\hat{u}_r\} := \{\hat{\Sigma^{xx}}\hat{v}_r/\sqrt{\hat{v}_r^\top \hat{\Sigma^{xx}}\hat{v}_r}\}$ (to read $\sigma_r(t)$ off $\bar{W}(t)$).

Step 3. Initialise $\bar{W}^{(0)} = V^{(0)} = \varepsilon \cdot I$ (aligned balanced; equivalent to $\bar{W}^{(0)} = \varepsilon \cdot U_x^\top U_x$ in the population eigenbasis).

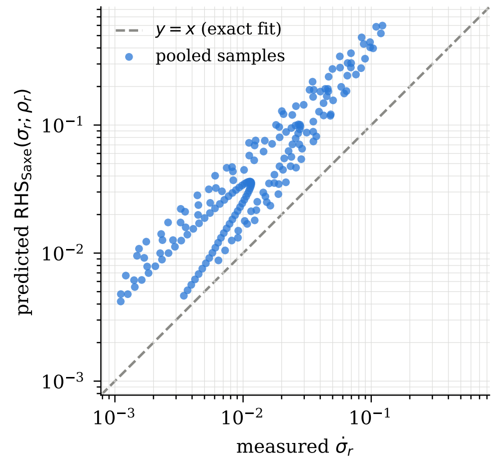
Step 4. Train: for $k = 0, \ldots, T_{\max} - 1$, $\bar{W}^{(k+1)} = \bar{W}^{(k)} - \eta \nabla_{\bar{W}} \mathcal{L}_n(\bar{W}^{(k)}, V^{(k)})$ with $\mathcal{L}_n = \frac{1}{2n}\|Y - V\bar{W}X\|_F^2$, identically for V; record $\sigma_r^{(k)} := \hat{u}_r^\top \bar{W}^{(k)}\hat{v}_r$.

Step 5. Plateau detection: stop at first k^* with $\max_t |\sigma_r^{(k^*)} - \sigma_r^{(k^*-50)}| < \tau_p$ for all r, or at $k^* = T_{\max}$.

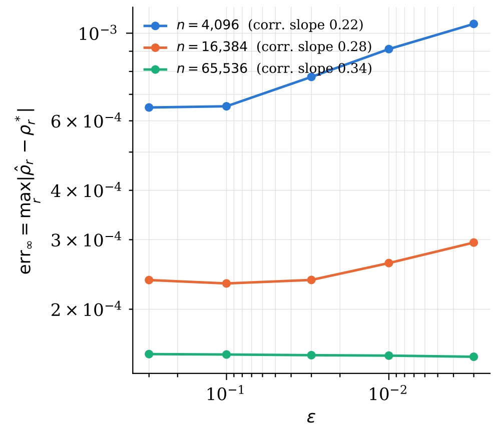
Step 6. For each r:

- If $\sigma_r^{(k^*)} > \eta_*(\varepsilon)$: $\hat{\rho}_r \leftarrow \sigma_r^{(k^*)L}$.
- Else, distinguish plateaued (zero branch) from still-decaying (negative branch) via $\sigma_r^{(k^*)}$ vs. $\sigma_r^{(k^*/2)}$ (tolerance not yet quantitatively derived — Remark 4.4 — reference code uses a heuristic threshold):
  - Plateaued: $\hat{\rho}_r \leftarrow 0$.
  - Still decaying: $\hat{\lambda}_r^{\text{neg}}$ from the late-time window via the difference quotient of Proposition 4.2's proof; extract $\hat{\mu}_r^{\text{cov}}$ from $\hat{\Sigma^{xx}}, \hat{\Sigma^{yx}}$; set $\hat{\rho}_r \leftarrow -\hat{\lambda}_r^{\text{neg}}/\hat{\mu}_r^{\text{cov}}$.

where $\eta_*(\varepsilon) := \varepsilon^{1/L}|\log \varepsilon|^2$.



(A)



(B)

Figure 1. **Left:** PlateauRecover, signed-spectrum recovery from a linear JEPA training run. Step 2 is the only call to sample covariances; Steps 3–6 (positive branch) operate on the trajectory alone. Cost $O(nd^2 + T_{\max}d^3)$; implementation in `experiments/plateau_recover_corrected.py` of the companion repository. **Right, top (a):** diagonal-ODE direction check (§5.2.2), $\varepsilon = 0.05$, $d = 6$, pooled over 191 in-motion samples across six features — points sit consistently above $y = x$ (median relative residual 0.74, max 0.86, normalised by $|\mathrm{RHS}_{\mathrm{Saxe}}|$): correct sign and order of magnitude, not a tight quantitative fit. **Right, bottom (b):** rate-isolation sweep (§5.2.3), $d = 10$, 80,000 steps, 3 seeds per point: $\mathrm{err}_\infty = \max_r |\hat{\rho}_r - \rho_r^*|$ vs. ε, one curve per n; annotated slope is the polynomial fit of $\log \mathrm{err}_\infty - \log|\log \varepsilon|$ against $\log \varepsilon$ (theory: 0.50). Both restricted to $L = 2$, positive branch.

5.2.1. *Setup.* In both experiments (with separate configurations) we generate synthetic (x, y) samples from a ground-truth pencil $(\Sigma^{yx}, \Sigma^{xx}) \in \mathbb{R}^{d\times d}$ with prescribed generalised eigenvalues $\rho_1^* > \rho_2^* > \cdots > \rho_d^*$ on the positive branch. We train a depth-$L = 2$ JEPA with aligned balanced initialisation $\bar{W}(0) = V(0) = \varepsilon \cdot I$ at the prescribed ε by full-batch SGD; plateau detection uses a rolling window with a relative-tolerance stopping rule. Experiment 2 (rate isolation) uses $d = 10$, the linear spectrum sweep $\rho_r^* = 1.0, 0.91, 0.82, \dots, 0.20$, learning rate 0.05, and up to 80,000 steps with a 50-step/5×10^{-4} plateau window (`experiments/plateau_recover_corrected.py` and `experiments/rate_isolation_probe.py` of the companion repo). Experiment 1 (ODE-form sanity check) uses a smaller $d = 6$ configuration with learning rate 0.02 for 40,000 steps (`experiments/ode_form_fit.py` of the companion repo). It only needs to verify the ODE's functional form, not exercise the full-scale sweep.

5.2.2. *Experiment 1: diagonal-ODE sanity check.* We first compare the trajectory's measured velocity against the diagonal ODE's predicted right-hand side, evaluated at the empirical $\sigma_r(t)$:

$$\mathrm{RHS}_{\mathrm{Saxe}}(\sigma;\rho) := L\mu_r\sigma^{2-1/L}(\rho-\sigma^L).$$

Figure 1a shows all six features with the correct sign. The residual is far larger than the ≈ 0.032 Proposition 2.5 predicts. We attribute this gap to an unresolved scaling constant we have not derived.

5.2.3. *Experiment 2: rate isolation.* This experiment checks whether the estimator's error shrinks at the theoretical rate as n grows:

$$\mathrm{err}_\infty = O(\varepsilon^{1/L}|\log\varepsilon|) + O_p(n^{-1/2}/\delta),$$

dynamics term plus sample-noise floor, $1/L = 0.5$ at $L = 2$. Figure 1b: error shrinks about $4\times$ from $n = 4{,}096$ to 65,536, tracking the predicted noise floor. Raw slopes move from -0.11 toward 0 as n grows, masking the underlying rate. Correcting for the log factor, slopes rise from 0.22 to 0.34, approaching the theoretical 0.50.

We attribute this to noise-floor saturation as predicted by Corollary 4.5. Pinning the corrected slope to within ±0.05 of 0.50 needs another order of magnitude beyond $n = 65{,}536$. We leave this and a real-data demonstration to follow-on work.

6. Related work

Joint Embedding Predictive Architecture (JEPA). LeCun [9] named the JEPA architecture, gave its two-encoder latent-prediction diagram, and laid out the collapse taxonomy later studies inherit — a self-supervised encoder trained without care can converge to trivial solutions, so anti-collapse mechanisms are a live engineering concern. Since then the field has moved toward using JEPA as a world model, e.g. Maes et al. [12]'s LeWorldModel.

Non-OLS methods for analysing regression structure. The pencil $(\Sigma^{yx}, \Sigma^{xx})$ this paper recovers is a classical object of study. Hotelling [7] introduced canonical correlation analysis on exactly this generalized eigenproblem. Closer to a dynamics-based claim, Varre et al. [19] characterize the $t \to \infty$ limit of gradient flow on a plain two-layer linear network as a min-norm interpolator — an ordering/timing result with no signed value and depth=2. Two recent JEPA-theoretic results also recover structure without training dynamics: Klindt et al. [8] prove identifiability of LeJEPA's learned representation at the population optimum up to an orthogonal transformation, while Cui et al. [4] bound downstream planning regret via a low-rank spectral factorization of an action-conditioned co-occurrence matrix, only once the encoder and predictor already sit at their population-optimal solution. None of the above use training dynamics as the recovery mechanism: the classical estimators require the full sample covariance (or co-occurrence) structure in hand, and Klindt et al./ Cui et al. characterize representations only once optimization has already finished. The present paper recovers the same kind of object from training dynamics alone, without ever forming $\hat{\Sigma^{xx}}, \hat{\Sigma^{yx}}$ directly.

Analysing regression structure via model training dynamics. A separate line of work asks whether training dynamics themselves reveal this structure, without a closed-form estimator. Littwin et al. [11] shows gradient-flow dynamics of a linear JEPA pair, with no added regularizer, preferentially learn high-regression-coefficient features faster, at critical time $\tilde{t}_r^* \asymp 1/(\lambda_r^* \varepsilon^{(2L-1)/L})$ — dynamics do encode regression structure, at a rate in the same $\varepsilon^{1/L}$-family as this paper's own $O(\varepsilon^{1/L}|\log\varepsilon|)$ recovery rate (§4). But the result assumes Σ^{xx} and Σ^{yx} share an eigenbasis (known in advance) and recovers a magnitude-linked critical time, not a signed coefficient - an assumption relaxed by Goh [6]. Dominé et al. [5] generalizes along a different axis — exact solutions across a continuous λ-balanced initialization spectrum, not only the zero-balanced case — but $\tilde{\Sigma^{xx}}, \tilde{\Sigma^{yx}}$ stay fixed and known throughout. Together these three confirm training dynamics genuinely encode regression structure but stop short on different axes. The present paper is the first to close all three at once, recovering an unknown, signed structure from dynamics alone.

Solving training dynamics. This paper's own proof strategy inherits its central diagonalization step from Saxe et al. [15, 16], who first showed that under aligned, orthogonal initialization a deep linear network's coupled gradient-flow ODEs decouple into per-mode scalar Bernoulli-type equations with closed-form sigmoidal trajectories; we diagonalize onto the generalized eigenbasis instead of an ordinary one, yielding the same Saxe-style depth exponents, but the decoupling step itself is classical and not re-derived here. Arora et al. [1, 2] formalize the "balanced initialization" condition this paper's own Assumption A2 restricts to, showing deep linear gradient descent is equivalent to a preconditioned single-layer system with a rigorous linear-rate convergence guarantee under approximately balanced initialization and whitened data — the optimization/convergence-rate side of the same initialization concept Saxe et al. solve exactly. Nam et al. [14] instead argues, independent of any new dynamics result, that layerwise-linear networks are the right minimal lens for training-dynamics phenomena generally, directly supporting this paper's own reason for studying the linear case first (§1). Together these establish that deep linear training dynamics are exactly solvable and worth studying on purpose even though OLS could already recover the population structure directly; none ask this paper's own question of that same machinery: identifiability of an unknown, signed structure.

Formal verification of learning theory. Machine-checked learning theory is a small but growing body of work. Bagnall and Stewart [3] started this line, formalizing PAC bounds in Coq for finite hypothesis classes. Sonoda et al. [17] made the major advance to infinite hypothesis classes, in Lean 4 against Mathlib — the same proof assistant this paper uses. A companion repository (`https://github.com/davidcagoh/jepa-rho-recovery`, public) contains the Lean development and the PlateauRecover reference implementation, alongside, not in place of, the paper's main mathematical contribution.

7. Discussion

Scope and Model Idealizations. Theorem 4.3 answers the question of the introduction sharply: linear JEPA's training trajectory contains all information needed to identify both the sign and (positive-branch) magnitude of ρ_r^*. Negative-branch magnitude is instead farmed out to sample covariance, a rate-optimal choice.

Linear-to-Non-Linear Transition. The argument is squarely in the linear, exact-gradient-flow regime. Three components break under non-linearisation, as in the LeWorldModel architecture of Maes et al. [12]. First, optimal V exists in closed form, which fails for non-linear predictors. Second, the generalised eigendecomposition of $(\Sigma^{yx}, \Sigma^{xx})$ is replaced by an unknown learned inner product (the encoder's pre-image of the predictor norm), making the analogue of ρ_r^* implicitly defined. Third, the diagonal-amplitude reduction is special to the linear case. Yet, establishing sufficient-statistic structure for the linear case is the prerequisite for asking which parts of the picture survive non-linearisation at all, and any answer will likely involve a learned metric on the encoder output rather than the fixed Σ^{xx} inner product. Within the linear setting, assumption A3's strictly separated discrete spectrum is unrealistic for high-dimensional inputs, where eigenvalues bunch under random projection, collapsing δ and degrading the basis-perturbation bound (13). A Marchenko and Pastur [13]-type density-of-states formulation is the natural fix, outside our present scope.

Technical Refinements & Open Analytical Problems. Two gaps remain inside the linear argument itself, both already disclosed at their point of use rather than new to this discussion. First, Proposition 2.5's residual constant C_R is not yet known to be ε-independent: the current derivation discharges it via a compactness argument. Pinning down an explicit constant needs a leading-order-cancellation argument for $V_{\mathrm{qs}}\bar{W}$ that has not been carried out (Appendix A). Second, Remark 3.2 shows the residual bound alone does not suffice to establish the sign trichotomy's case (ii) ceiling (Theorem 3.1) at $L = 2$ — the one depth tested empirically. (This does not show the ceiling is actually violated in practice.) A third, downstream gap follows from the second: Remark 4.4 notes that no finite-time, noise-robust decision rule yet separates the zero and negative branches. This is because the qualitative ceiling such a rule would rely on is itself unresolved at $L = 2$. Machine-checking via Lean (§6) rules out logical gaps and forgotten edge cases in the stated theorems. It does not certify that the definitions are the right ones, that the assumptions are natural, or that the result is interesting, and we do not treat it as doing so. The two named axioms in Appendix A are the places where the Lean chain stops and ordinary mathematical trust in a cited result takes over. The first gap above is an incomplete derivation internal to this paper's own argument, marked explicitly (Appendix A, †). The second is its own separately named row in Appendix A's theorem-to-Lean crosswalk table.

Empirical Extensions & Future Applications. Some experimental results diverge from what the theory alone predicts, and several natural empirical extensions remain. The diagonal-ODE sanity check's residual (§5.2.2) sits well above Proposition 2.5's asymptotic bound; isolating the constant separating full-batch loss normalisation from continuous-time gradient-flow rescaling remains unresolved. Whether $\{\hat{\rho}_r\}$ predicts downstream-task ordering the way the dynamics-ordering theorem predicts feature-learning ordering is likewise open. PlateauRecover still awaits a real, non-synthetic representation-learning benchmark — training a small JEPA from random initialisation and reading off $\hat{\rho}_r$ directly. The experiments reported here (§5.2) also cover only the positive branch, leaving the zero and negative branches, the sign trichotomy of §3, and the decay-to-zero classifier of §4.2 for later empirical work, despite signed recovery being a headline contribution. A direct comparison against generalised-eigenvalue regression at fixed n would locate the widest sample-efficiency gap between the two approaches.

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Appendix A. Lean verification catalogue

The Lean 4 development for this paper is in the repository `https://github.com/davidcagoh/jepa-rho-recovery` (public) and builds against Mathlib v4.28.0 (commit `7e01a1b`). Items marked "(companion paper)" below cite Goh [6]'s own Lean development, also public, at `https://github.com/davidcagoh/jepa-learning-order`. The crosswalk below maps each numbered theorem in the paper to its Lean target. "Sorry-free" means the theorem body contains no `sorry`; this is independent of axiom use — a sorry-free result can still depend on a named axiom as an explicit hypothesis, disclosed in the Status column.

Paper	**Lean target**	**Status**
Prop 2.4 (quasi-static)	`QuasiStatic.quasiStatic_rigorous`	sorry-free
Prop 2.5 (diagonal ODE)	`Saxe.diagAmp_ODE` (companion paper)	sorry-free[†]
Prop 4.1 (plateau estimator)	`Saxe.rho_hat_plateau_rate`; `Saxe.signed_recovery_pos_magnitude_plateau`; `Saxe.sigma_positive_branch_converges`	sorry-free (A2)[‡]
Thm 3.1 (sign trichotomy), cases (i)–(ii)	`Saxe.sigma_positive_branch_converges`; `SignedODE.sigma_zero_branch_constant`	sorry-free
Thm 3.1, case (iii)	not cited[§]	argued directly
Rmk 3.2 (L=2 ceiling gap)	`ZeroBranchResidual.zero_branch_L2_ceiling_violation`	sorry-free
Cor 3.3 (sign identification)	`SignedRecovery.sign_identification_pos_forward`; `SignedRecovery.sign_identification_zero_forward`; `SignedRecovery.sign_identification_neg_forward`	sorry-free
Prop 4.2 (negative-branch λ_r^*-rate)	not cited[§]	argued directly
Thm 4.3 (signed decomposition)	`Main.signed_decomposition`[¶]	sorry-free
Cor 4.5 (finite-sample rate)	`FiniteSample.end_to_end_rate`[¶]	sorry-free (A1)

[†] C_R is pinned via a compactness argument, not shown ε-independent (§7). [‡] Restricted to $\rho_r^* > 0$, dynamics term only (no sample-noise floor); the Lean proof uses qualitative convergence plus a topological existence argument for T_*, mechanically different from the quantitative Grönwall bound in §4.1 but giving the same rate. [§] The negative-branch Lean theorems assume the pre-correction (deprecated) ODE form, not this paper's (8), and have not been re-verified against it, so both results are argued directly in the main text instead. [¶] Both extended additively from earlier positive-branch-only versions to cover all branches, composed from already-proven pieces via the same Delta-method hypothesis-bundling style throughout; no previously-verified theorem touched, no new axiom.

Named axioms. The full development declares two named axioms; one is load-bearing on the finite-sample chain and the other on the dynamics chain.

(A1) `Concentration.matrix_bernstein_subgaussian` — the matrix Bernstein inequality of Tropp [18, Thm 1.6.2], quoted per the CompCert convention [10]. Used by Corollary 4.5. *Why axiomatised:* Mathlib has scalar concentration inequalities but not, at the time of writing, an operator-norm matrix-Bernstein bound in directly usable form; packaging Tropp [18]'s proof is a Mathlib-engineering project, not a result we chose to skip despite it being easy.

(A2) `Saxe.saxe_exact_solution_exists` — Picard–Lindelöf existence for the Saxe-form Bernoulli ODE $\dot{\sigma} = L\mu\sigma^{2-1/L}(\rho - \sigma^L)$ on a compact interval. Standard scalar-ODE existence; quoted per CompCert. Used by Proposition 4.1 and downstream. *Why axiomatised:* Mathlib has general Picard–Lindelöf machinery for locally Lipschitz vector fields, but instantiating it for this specific nonlinear scalar form together with the hitting-time reachability clause is, again, an engineering gap rather than a deliberate shortcut — the companion paper's own open-problems discussion (Goh 6, §Open problems) names exactly this as future Mathlib-engineering work.

Appendix B. Deferred proofs

Proofs deferred from the main text: the off-diagonal bound and positive-definite lower bound used in §2.1, the coupled gradient-flow system and quasi-static decoder bound used in §2.2 (all technical, but with a role already clear from the main-text statements), and the finite-sample perturbation bounds summarised in §4.3.

B.1. **Off-diagonal bound (Lemma 2.1).**

Proof. This is the fundamental-theorem-of-calculus argument of Goh [6, §6, Prop. (Bootstrap consistency), item 1] (`offDiag_ftc`, `BootstrapLemmas.lean`, companion repository, public): c_{rs} is linear in $\bar{W}$, so Assumption A4(i)'s bound $\|\dot{\bar{W}}(t)\|_F \le K_1\varepsilon^2$ integrates directly to $|c_{rs}(t)| \le C'\varepsilon^{1/L}$ on $[0, t^*_{\max}]$, using $c_{rs}(0) = 0$ under (3). The argument is sign-agnostic in ρ^*_r, so it transfers unchanged from the companion's all-positive setting. □

B.2. **Positive-definite lower bound (Corollary 2.2).**

Proof. This is the Gershgorin argument of Goh [6, §6, Prop. (Bootstrap consistency), item 2] (`pd_lower_from_offDiag`, companion repository, public): diagonal dominance $C'(d-1) < c_w$ forces $\bar{W}(t)$ invertible via Gershgorin's theorem, giving $\sigma_{\min}(\bar{W}(t)) \ge (c_w - C'(d-1))\varepsilon^{1/L}/\sqrt{d}$ and hence $\bar{W}\Sigma^{xx}\bar{W}^\top \succeq c_0\varepsilon^{2/L} I$ using $\Sigma^{xx} \succ 0$. Sign-agnostic, as above. □

B.3. **Coupled gradient-flow system (Lemma 2.3).**

Proof. (5): direct differentiation of $\mathcal{L} = \frac{1}{2}\mathbb{E}\|y - V\bar{W}x\|_2^2$ under the expectation, using $\Sigma^{xx} = \mathbb{E}[xx^\top]$, $\Sigma^{yx} = \mathbb{E}[yx^\top]$.

(6): the identical balanced-flow reduction is derived in Goh [6, §2.3] (companion repository, public) from the same balance condition (Assumption A2) and Arora et al. [1]; not re-derived here. □

B.4. **Quasi-static decoder, rigorous (Proposition 2.4).**

Proof. The identical Grönwall tracking argument is proved in Goh [6, §6, Prop. (Bootstrap consistency), item 3] (`tracking_bound_from_gronwall`, assembled from `contraction_ode_structure` and `contractive_gronwall_decay`, companion repository, public), under the same hypotheses (Assumption A4(i), Corollary 2.2): writing $\Delta V := V - V_{\text{qs}}(\bar{W})$, $\dot{\Delta V} = -\Delta V\, \bar{W}\Sigma^{xx}\bar{W}^\top - \dot{V}_{\text{qs}}(\bar{W})$ contracts at rate $c_0\varepsilon^{2/L}$ against a drift bounded by $D_0\varepsilon^2$, giving $\|\Delta V(t)\|_F \le C_{\text{qs}}\varepsilon^{2(L-1)/L}$. Independently verified in our own Lean development (`QuasiStatic.quasiStatic_rigorous`, sorry-free). Sign-agnostic. □

B.5. **Matrix-Bernstein perturbation.** Under sub-Gaussian assumptions on (x, y) with constants σ_x, σ_y, Tropp [18, Thm 1.6.2] gives

$$\mathbb{P}\Big[\big\|\hat{\Sigma^{xx}} - \Sigma^{xx}\big\|_{\text{op}} > t\Big] \le 2d\exp\left(-\frac{nt^2}{2\sigma_x^4 + 2\sigma_x^2 t/3}\right), \tag{12}$$

and analogously for $\hat{\Sigma^{yx}} - \Sigma^{yx}$; taking $t = c\sqrt{\sigma_x^4\log(d/\nu)/n}$ gives $\|\hat{\Sigma^{xx}} - \Sigma^{xx}\|_{\text{op}} \le c\sqrt{\sigma_x^4\log(d/\nu)/n}$ with probability at least $1-\nu$.

B.6. **Davis–Kahan / Wedin for the generalised eigenproblem.** A Wedin-type bound converts the covariance perturbation into an eigenvector perturbation:

$$\sin\angle\big(\hat{v}_r, v^*_r\big) \;\le\; \frac{\|\hat{\Sigma^{yx}} - \Sigma^{yx}\|_{\text{op}} + |\rho^*_r|\,\|\hat{\Sigma^{xx}} - \Sigma^{xx}\|_{\text{op}}}{\delta_r - O(\|\hat{\Sigma^{xx}} - \Sigma^{xx}\|_{\text{op}})} \;\le\; \frac{C_W}{\delta}\sqrt{\frac{\log(d/\nu)}{n}} \tag{13}$$

on the event of (12), with C_W depending on σ_x, σ_y, $\|\Sigma^{xx}\|, \|\Sigma^{yx}\|$ — standard but technical for the generalised (non-symmetric) eigenproblem, formalised as `SampleNoise.sample_eigenvalue_perturbation` in the companion repo.

Department of Computer Science, University of Toronto
Email address: `daveed@cs.toronto.edu`